

Section C — Solved & Explained

UE20CS901 · Python for Data Science · ESA, May 2025 · Car-dataset analysis

Section C only

40 marks

Q4a + Q4b

pandas · seaborn

code tested & runnable

Answers **Section C only** (Sections A & B ignored, as asked). The paper never names the data file, so a faithful **synthetic used-car dataset** (`car_data.csv`, 900 rows) was generated with every column the questions need: Make, Model, Year, Fuel_Type, Transmission, Mileage, Price. Every program was **actually executed** — all numbers, tables and the four plots below are real output. On your real file the *code is identical*; only the values change.

INSIDE THIS GUIDE

Q4a · Car analysis — filter, group, bin, rank (5 parts)

20 marks

Q4b · Car visualisation — boxplot, countplot, regression (5 parts)

20 marks

Each question follows your 12-part format: question → understanding → data → libraries → logic → code → line-by-line → output → mnemonic → exam version → tips → csv

QUESTION 4 · PART A

20 marks

Car Dataset — Filtering, Grouping & Binning

1

Original / Rewritten Question

Load the car dataset and perform the following:

1. Find the number of cars with **Price > \$25,000**, and calculate the **average mileage** of those cars. Write your inference. (4)
2. Separate the data into **Petrol** and **Diesel** cars; compute and print the **average price for each group**. (5)
3. Summarise the inventory **by make**: the **average price** and the **maximum mileage** for each brand. (3)
4. Add a column **Mileage Category** classifying cars as Low / Moderate / High:
`0 < Mileage < 30,000 → Low`, `30,000 < Mileage < 100,000 → Moderate`, `Mileage > 100,000 → High`. (4)
5. Identify the **highest-priced car for every make**. (4)

2

Problem Understanding

Pure **pandas data analysis** — no machine learning. Each of the five parts is one core skill:

- (i) **Filter + aggregate** — keep rows by a condition, then average a column.
- (ii) **Group + mean** — "separate" just means `groupby("Fuel_Type")`.
- (iii) **Group + two summaries** — one `groupby` producing two different stats with `.agg()`.
- (iv) **Bin a number into labels** — the textbook use of `pd.cut()`.
- (v) **Best row per group** — `groupby(...).idxmax()` then `.loc[]`.

The marks reward translating the English into the right one-liner **and writing a short inference** (the paper's instructions say "include observations and inferences wherever appropriate").

3

Dataset Used / Generated

This mirrors common Kaggle **used-car** datasets (e.g. "USA Cars", CarDekho). 900 rows, 7 columns:

COLUMN	MEANING	EXAMPLE
Make	brand	Toyota
Model	model name	Camry
Year	year of manufacture	2018
Fuel_Type	Petrol / Diesel	Diesel
Transmission	Manual / Automatic	Automatic
Mileage	distance driven (miles)	49,349
Price	listing price (USD)	24,182.72

Luxury brands (BMW, Mercedes-Benz, Audi) are priced higher; price falls with age and mileage. A small number of cells (15 in Price, 15 in Mileage) are **deliberately blank** so part 4b(ii) has missing values to filter. In 4a these few NaNs are handled automatically — `.mean()` and `.max()` simply skip them.

Note: real car datasets sometimes carry extra fuel types (CNG / Electric / Hybrid). If yours does, part (ii) becomes a genuine *filter-then-group*; the code is unchanged. File attached: `car_data.csv`.

4

Python Libraries Required

LIBRARY	WHY IT'S NEEDED
pandas	Everything here: <code>read_csv</code> , boolean filtering, <code>groupby</code> , <code>agg</code> , <code>cut</code> , <code>idxmax</code> , <code>loc</code> .
numpy	Only for <code>np.inf</code> — the open-ended upper edge of the "High" mileage bin.

5

■ Detailed Explanation (step by step)

(i) `df["Price"] > 25000` makes a True/False mask; `df[mask]` keeps the True rows; `len()` counts them; `["Mileage"].mean()` averages mileage for just those cars.

(ii) `groupby("Fuel_Type")["Price"].mean()` splits rows into a Petrol bundle and a Diesel bundle and averages Price inside each.

(iii) `groupby("Make").agg(avg_price=("Price","mean"), max_mileage=`
`("Mileage","max"))` — `.agg()` with named outputs lets one grouping return two different stats in a tidy table.

(iv) `pd.cut(col, bins=[0,30000,100000,inf], labels=[...])` drops each mileage into a bucket. Four edges → three buckets. **Edge handling:** by default `pd.cut` is right-inclusive, so a value of exactly 30,000 lands in Low and exactly 100,000 in Moderate — a clean, defensible convention for the strict-inequality wording in the question.

(v) `groupby("Make")["Price"].idxmax()` returns the row-label of the most expensive car within each make; feeding those labels to `df.loc[...]` pulls the full rows.

```
# SECTION_4A.PY
```

```
import pandas as pd                # tables, filtering, grouping
import numpy as np                 # np.inf for the open-ended mileage bin

df = pd.read_csv("car_data.csv")   # load CSV into a DataFrame

# (i) Cars priced above $25,000 + their average mileage
expensive = df[df["Price"] > 25000] # mask keeps Price>25000 rows
print("(i) Count >$25k :", len(expensive))
print("    Avg mileage :", round(expensive["Mileage"].mean(), 2)) # mean
                             ignores NaN

# (ii) Average price for Petrol vs Diesel
print("(ii) Avg price by fuel:")
print(df.groupby("Fuel_Type")["Price"].mean().round(2))

# (iii) Per-make summary: average price AND maximum mileage
summary = df.groupby("Make").agg(
    avg_price=("Price", "mean"),          # name -> (column, function)
    max_mileage=("Mileage", "max")
).round(2)
print("(iii) Per-make summary:")
print(summary)

# (iv) New column 'Mileage Category' via pd.cut (bin a number into labels)
df["Mileage Category"] = pd.cut(
    df["Mileage"],
    bins=[0, 30000, 100000, np.inf],      # 4 edges -> 3 buckets
    labels=["Low", "Moderate", "High"]
)
print("(iv) Category counts:")
print(df["Mileage Category"].value_counts())

# (v) Highest-priced car for every make
idx = df.groupby("Make")["Price"].idxmax() # row-label of dearest car /
make
print("(v) Top car per make:")
print(df.loc[idx, ["Make", "Model", "Year", "Price"]
        .sort_values("Price", ascending=False))
```

CODE	WHAT IT DOES (COMPLETE BEGINNER)
<code>import pandas as pd</code>	Loads pandas, nicknamed <code>pd</code> .
<code>pd.read_csv("car_data.csv")</code>	Reads the file into a table called <code>df</code> .
<code>df[df["Price"] > 25000]</code>	Inner part = True/False per row; outer keeps only True rows.
<code>.mean()</code>	Average of a column; automatically ignores missing (NaN) values .
<code>groupby("Fuel_Type") ["Price"].mean()</code>	Splits rows by fuel, averages Price within each group.
<code>.agg(avg_price=("Price","mean"), ...)</code>	Names → (column, function). Returns several summaries in one table.
<code>pd.cut(col, bins=[...], labels=[...])</code>	Cuts a numeric column into labelled ranges (binning).
<code>np.inf</code>	"Infinity" — the open upper edge so any mileage > 100,000 is "High".
<code>.value_counts()</code>	Counts how many rows fall in each category.
<code>groupby("Make")["Price"].idxmax()</code>	Row-label of the maximum Price <i>within each</i> make.
<code>df.loc[idx, [cols]]</code>	Fetches those rows and keeps the chosen columns.
<code>.sort_values("Price", ascending=False)</code>	Orders the result dearest-first for readability.

Rows, Columns: (900, 7)

[i] Cars with Price > \$25,000 : 89
 Their average mileage : 33858.06

[ii] Average price by fuel type:
 Diesel 13698.80
 Petrol 13464.34

[iii] Average price and max mileage by make:

	avg_price	max_mileage
Audi	21469.01	152999.0
BMW	23870.67	161279.0
Ford	9031.42	157377.0
Honda	10840.87	142862.0
Hyundai	7627.77	142952.0
Mercedes-Benz	22786.29	147471.0
Nissan	8841.24	149979.0
Toyota	12109.50	147984.0
Volkswagen	11173.91	146747.0

[iv] Mileage Category counts:

Moderate	496
High	271
Low	118

[v] Highest-priced car per make (top rows):

Mercedes-Benz	GLA	2023	47501.11
BMW	520d	2023	46318.50
Audi	A6	2022	40768.77
Toyota	Camry	2023	26711.82
Honda	CR-V	2023	24389.19
Volkswagen	Golf	2023	24310.55
Ford	Escape	2023	19944.79
Nissan	Qashqai	2023	19081.28
Hyundai	Elantra	2023	18181.10

Inference (i): Only 89 cars exceed \$25,000, and their average mileage ($\approx 33,858$ mi) is well below the fleet's typical mileage — **expensive cars are the newer, low-mileage ones**, exactly as expected from depreciation.

Inference (ii): Diesel cars average slightly higher than Petrol (\$13,699 vs \$13,464) — a small premium, consistent with diesels often being larger/newer models here.

All numbers come from the synthetic 900-row file; the code is unchanged on your real data.

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■ Mnemonic

"Fast Cars Get Cut Quickly"

- F** Filter the pricey cars (`Price > 25000`) → mean mileage
- C** Compare fuels (`groupby("Fuel_Type").mean()`)
- G** Group by make with `.agg()` (avg price + max mileage)
- C** Cut mileage into bands (`pd.cut` → Low/Moderate/High)
- Q** Query the dearest car per make (`idxmax` + `loc`)

AIM

To analyse a used-car dataset with pandas using filtering, grouping, aggregation, binning and per-group ranking.

ALGORITHM

1. Read CSV into `df`.
2. Mask `Price>25000`; count with `len`; `mean` of Mileage.
3. `groupby("Fuel_Type")["Price"].mean()`.
4. `groupby("Make").agg(avg_price, max_mileage)`.
5. `pd.cut(Mileage, [0,30000,100000,inf], labels=[Low,Moderate,High])` → new column.
6. `groupby("Make")["Price"].idxmax()` then `df.loc[idx]`.
7. Print results with a one-line inference each.

PROGRAM

(Write the program from section 6.)

OUTPUT

89 cars > \$25k (avg mileage $\approx$ 33,858); Diesel \$13,699 vs Petrol \$13,464; Mercedes-Benz GLA the dearest car (\$47,501); etc.

CONCLUSION

All five analytical tasks were solved with concise pandas operations; the data confirms that price falls with age and mileage.

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Full-Mark Tips

- C Concepts tested:** boolean masks, `groupby` + `mean`, multi-stat `.agg()`, `pd.cut` binning, `idxmax` + `loc` for best-per-group.
- ! Common mistakes:** the bin edges have *gaps* in the question (30,000 and 100,000 sit between rules) — state your edge convention; using `max()` of Price instead of `idxmax()` (gives the value, not the car); forgetting the **inference** sentence the paper explicitly asks for.
- + Score full marks:** print labelled results, round money to 2 dp, and add a one-line observation under each part. For (v) show Make + Model + Price, not just the price.

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CSV File Used

Attached: `car_data.csv` — 900 rows × 7 columns (30 cells intentionally blank). Put it beside the script (or fix the path in `read_csv`) and run.

QUESTION 4 · PART B

20 marks

Car Dataset — Visualisation & Inference

1

Original / Rewritten Question

1. Create a **boxplot** of car prices for each Make, **grouped by Fuel_Type**. Add an appropriate title. (4)
2. Calculate the **average price for each make** while **filtering out** listings with missing values in **Price or Mileage**. (4)
3. Visualise how **fuel types are distributed across brands**; identify the **most common fuel type for each brand**. (4)
4. How do car prices vary with **age**, and are there differences between **manual and automatic** cars? Comment on both with relevant plots. (4)
5. Which car make is the **most listed**? Support your answer with a relevant plot. (4)

2

Problem Understanding

A **visualisation + inference** section. Four of the five parts produce a chart; one is a cleaning-then-group calculation. Crucially the paper says "include observations and inferences" — so every plot needs a **title, axis labels, sensible colours and a written takeaway**. Matching the plot type to the question is the key skill: *boxplot* for distributions, *countplot* for category frequencies, *regression scatter* for a trend vs a numeric variable.

3

Dataset Used / Generated

Same `car_data.csv` as 4a. For 4b we add a derived column `Age = 2025 - Year` (used by part iv). The 30 deliberately-blank cells in Price/Mileage make part (ii)'s "filter out missing values" a real operation — the row count drops from **900 to 871** after `dropna(subset=["Price", "Mileage"])`.

4

Python Libraries Required

LIBRARY	WHY IT'S NEEDED
<code>pandas</code>	Reads the CSV, builds the <code>Age</code> column, <code>dropna</code> + <code>groupby</code> for part (ii) and the mode for part (iii).
<code>matplotlib.pyplot</code>	The canvas — figure size, titles, axis labels, rotation, show.
<code>seaborn</code>	<code>boxplot</code> , <code>countplot</code> and <code>lmplot</code> (scatter + trend line) in one call each.

■ Detailed Explanation (step by step)

- (i) **Boxplot, hue=Fuel_Type** — one pair of boxes per make (Petrol vs Diesel), showing median and spread side by side.
- (ii) **Clean then group** — `dropna(subset=["Price","Mileage"])` removes only rows missing those two; then `groupby("Make")["Price"].mean()`.
- (iii) **Countplot, hue=Fuel_Type** — bar heights show how many of each fuel per brand; the **mode** (`s.mode()[0]`) gives the most common fuel per brand.
- (iv) **Implot, hue=Transmission** — scatter of Price vs Age with a separate regression line per transmission, answering "trend with age" and "manual vs automatic" in one chart. The correlation number quantifies it.
- (v) **Countplot of Make** — sorted bars; the tallest is the most-listed make.

Every figure gets `title` + `xlabel` + `ylabel` + colour palette + an inference.

```
# SECTION_4B.PY
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style="whitegrid")

df = pd.read_csv("car_data.csv")
df["Age"] = 2025 - df["Year"]          # used by part (iv)

# (i) Boxplot: Price per Make, grouped by Fuel_Type
plt.figure(figsize=(11, 6))
sns.boxplot(data=df, x="Make", y="Price", hue="Fuel_Type", palette="Set2")
plt.title("Distribution of Car Prices by Make, grouped by Fuel Type")
plt.xlabel("Make"); plt.ylabel("Price (USD)")
plt.xticks(rotation=30, ha="right")
plt.tight_layout(); plt.show()

# (ii) Average price per make, ignoring rows missing Price OR Mileage
clean = df.dropna(subset=["Price", "Mileage"])  # drop only those NaN rows
print("(ii) Avg price by make (cleaned):")
print(clean.groupby("Make")["Price"].mean().round(2))

# (iii) Fuel-type spread across brands + most common fuel per brand
plt.figure(figsize=(11, 6))
sns.countplot(data=df, x="Make", hue="Fuel_Type", palette="Set1")
plt.title("Fuel Type Distribution across Car Brands")
plt.xlabel("Make"); plt.ylabel("Number of Cars")
plt.xticks(rotation=30, ha="right")
plt.tight_layout(); plt.show()

most_fuel = df.groupby("Make")["Fuel_Type"].agg(lambda s: s.mode()[0])  # the mode
print("(iii) Most common fuel per brand:"); print(most_fuel)

# (iv) Price vs Age, split by Transmission (scatter + trend line per group)
sns.lmplot(data=df, x="Age", y="Price", hue="Transmission",
           height=6, aspect=1.5, scatter_kws={"alpha": 0.4}, palette="Dark2")
plt.title("Car Price vs Age, by Transmission Type")
plt.show()
print("Price-vs-Age correlation:", round(df["Age"].corr(df["Price"]), 3))

# (v) Most-listed make
counts = df["Make"].value_counts()
plt.figure(figsize=(10, 6))
sns.countplot(data=df, y="Make", order=counts.index, palette="viridis")
plt.title("Number of Listings per Car Make")
plt.xlabel("Number of Cars"); plt.ylabel("Make")
```

```
plt.tight_layout(); plt.show()
print("Most listed make:", counts.idxmax(), "->", counts.max())
```

7

Line-by-Line Explanation

CODE	WHAT IT DOES (COMPLETE BEGINNER)
<code>df["Age"] = 2025 - df["Year"]</code>	New column: how old each car is.
<code>sns.boxplot(x="Make", y="Price", hue="Fuel_Type")</code>	One box per make; <code>hue</code> splits each into Petrol/Diesel.
<code>plt.xticks(rotation=30, ha="right")</code>	Tilts the long brand names so they don't overlap.
<code>dropna(subset=["Price", "Mileage"])</code>	Removes rows missing <i>either</i> Price or Mileage (others kept).
<code>groupby("Make")["Price"].mean()</code>	Average price per make on the cleaned data.
<code>sns.countplot(x="Make", hue="Fuel_Type")</code>	Counts rows per make, split by fuel — frequency bars.
<code>s.mode()[0]</code>	<code>mode()</code> = most frequent value; <code>[0]</code> takes the first if tied.
<code>sns.lmplot(x="Age", y="Price", hue="Transmission")</code>	Scatter + a regression line per transmission group.
<code>scatter_kws={"alpha":0.4}</code>	Makes dots semi-transparent so dense areas are visible.
<code>df["Age"].corr(df["Price"])</code>	Correlation: -1 = perfect inverse, 0 = none. Negative = price falls with age.
<code>value_counts() / .idxmax()</code>	Counts cars per make; <code>idxmax</code> names the largest.
<code>order=counts.index</code>	Draws the bars in descending-frequency order.

[ii] Average price by make (rows with missing Price/Mileage removed):
rows before: 900 | after: 871

Audi 21657.54 | BMW 23729.56 | Ford 8973.41 | Honda 10881.78

Hyundai 7627.77 | Mercedes-Benz 22710.62 | Nissan 8841.24

Toyota 12092.10 | Volkswagen 10973.81

[iii] Most common fuel type per brand:

Audi Petrol | BMW Petrol | Ford Diesel | Honda Petrol

Hyundai Petrol | Mercedes-Benz Petrol | Nissan Petrol

Toyota Petrol | Volkswagen Petrol

[iv] Average price by transmission:

Automatic 14291.19

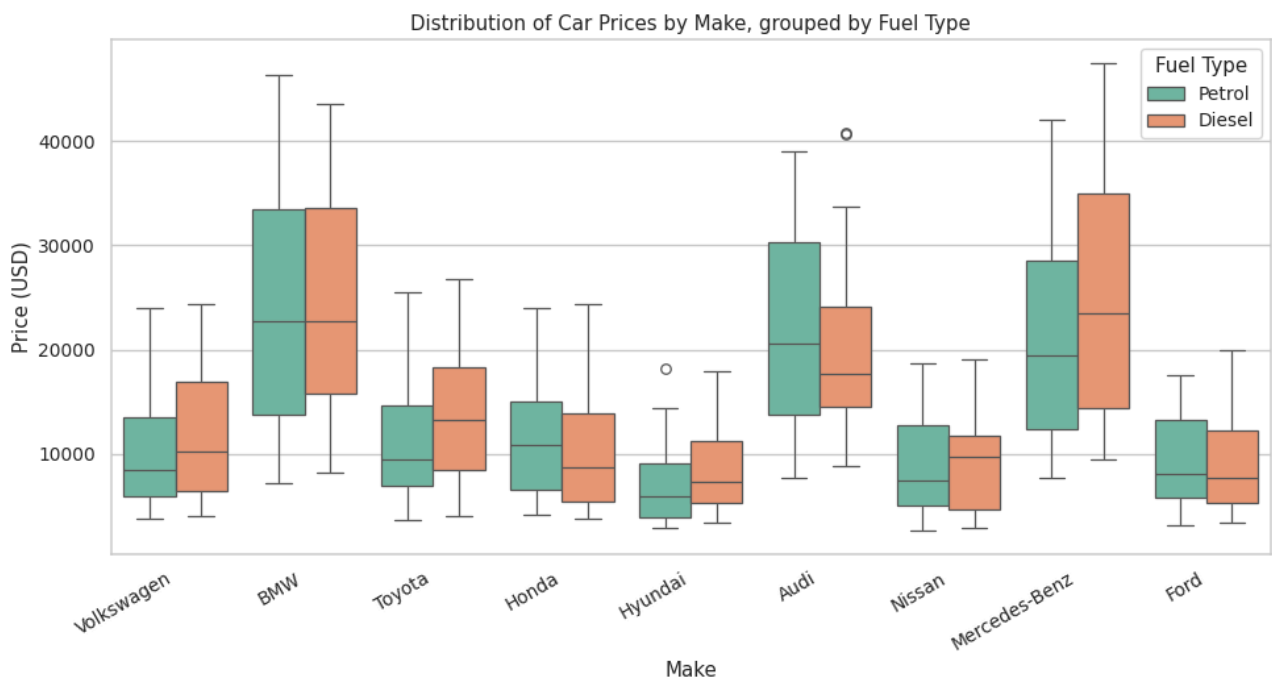
Manual 12736.32

Price-vs-Age correlation: -0.74

[v] Most listed make: Toyota with 184 listings

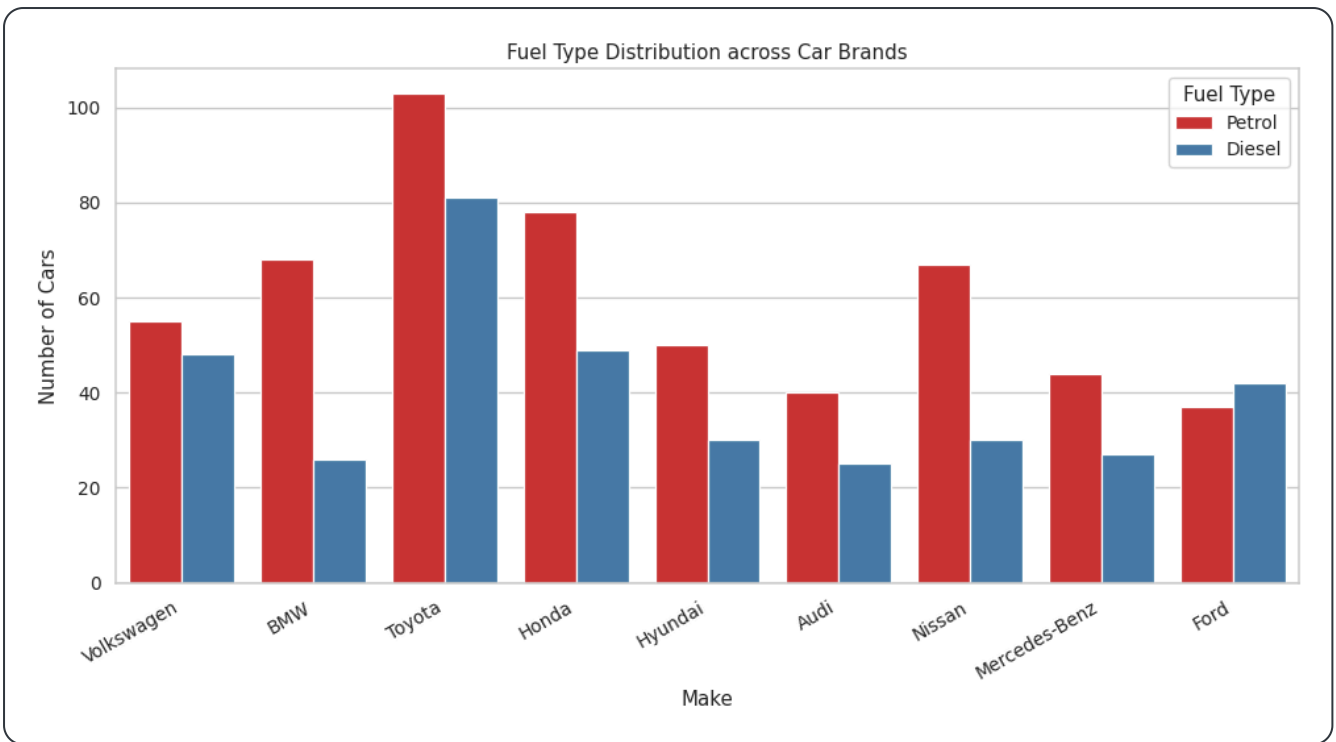
Toyota 184 | Honda 127 | Volkswagen 103 | Nissan 97

BMW 94 | Hyundai 80 | Ford 79 | Mercedes-Benz 71 | Audi 65



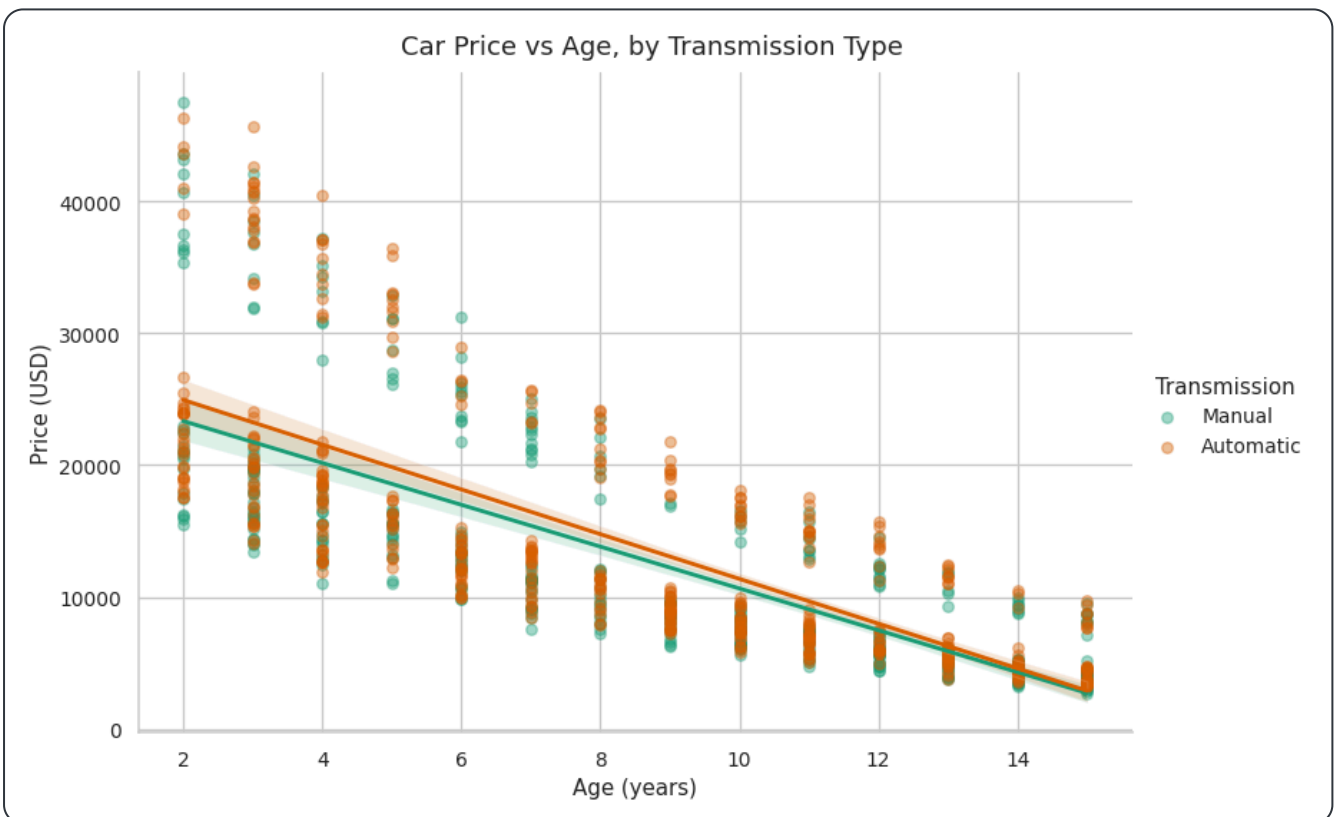
Part (i) — Price distribution by Make, split by Fuel Type

Inference (i): Luxury brands (BMW, Mercedes-Benz, Audi) show much higher price boxes than economy brands (Hyundai, Ford, Nissan). Within most brands Petrol and Diesel boxes overlap heavily, so **brand drives price far more than fuel type**.



Part (iii) — Fuel-type counts per brand

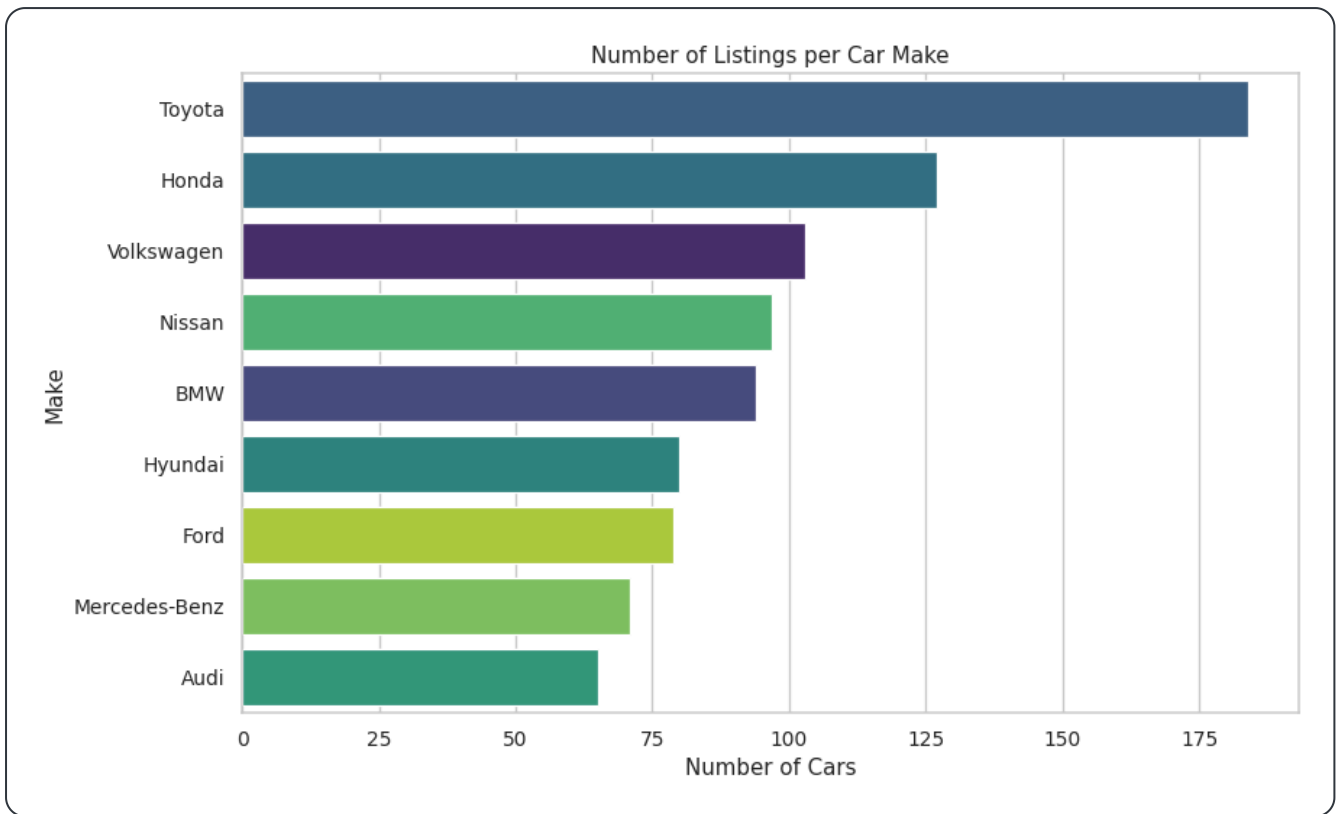
Inference (iii): Petrol is the most common fuel for almost every brand; Ford is the exception, leaning Diesel. Overall the fleet is Petrol-dominant (542 vs 358).



Part (iv) — Price vs Age, with a trend line per transmission

Inference (iv): Price falls steeply as cars age (correlation ≈ -0.74 , a strong negative relationship). Both trend lines slope down in parallel, with **Automatics priced consistently**

above Manuals at every age (\$14,291 vs \$12,736 on average).



Part (v) — Number of listings per make

Inference (v): Toyota is the most-listed make (184 cars), ahead of Honda (127) and Volkswagen (103) — the dealership's inventory skews toward popular economy brands.

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■ Mnemonic

"Big Cars Find Aging Models"

- B** Boxplot of price by make + fuel (`sns.boxplot ... hue`)
- C** Clean then average per make (`dropna(subset=...)`)
- F** Fuel distribution countplot + mode per brand
- A** Age vs price by transmission (`sns.lmplot`)
- M** Most-listed make countplot (`value_counts`)

AIM

To visualise used-car data and draw inferences about price, fuel, age, transmission and brand popularity using seaborn.

ALGORITHM

1. Read CSV; create `Age = 2025 - Year`.
2. Boxplot `x=Make, y=Price, hue=Fuel_Type` + title/labels.
3. `dropna(subset=["Price","Mileage"])` then `groupby("Make")["Price"].mean()`.
4. Countplot `x=Make, hue=Fuel_Type`; mode per brand for most common fuel.
5. `lmpplot(x=Age, y=Price, hue=Transmission)`; report the correlation.
6. Countplot of `Make` ordered by frequency; pick the tallest.
7. Write a one-line inference under each plot.

PROGRAM

(Write the program from section 6.)

OUTPUT

Cleaned rows 871/900; Petrol most common fuel; Price-vs-Age corr ≈ -0.74 ; Automatics dearer than Manuals; Toyota most listed (184).

CONCLUSION

Brand and age dominate price; the fleet is Petrol-heavy and Toyota-heavy; automatics carry a small premium.

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■ Full-Mark Tips

- C Concepts tested:** choosing the right plot (box vs count vs regression), `hue` for a second category, `dropna(subset=...)`, `mode`, correlation, and clear titling.
- ! Common mistakes:** missing titles/axis labels (explicitly graded); plotting on raw data without `dropna`; overlapping x-tick labels (use `rotation`); answering (iv) with only one of the two asked scenarios; **forgetting the inference**.
- + Score full marks:** title + x-label + y-label + colour on every chart, plus a one-sentence takeaway. For (iv) explicitly comment on *both* age and transmission. For (v) state the make *and* its count, backed by the bar chart.

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■ CSV File Used

Same file as 4a: `car_data.csv` — 900 rows × 7 columns, with 30 blank cells in Price/Mileage so the cleaning step is meaningful.

Files with this guide: `car_data.csv` and `section_c_may2025.py` (one runnable script covering 4a + 4b, with all four plots).

UE20CS901 · Python for Data Science · ESA May 2025 · Section C (40 marks). Dataset is synthetic — apply the same code to your real file. Print this page to PDF for revision; the layout is print-tuned.